Name:_____ Period:_____
Earth and Space Science

CYCLES AND SYSTEM MODELS
LEARNING GUIDE

Learning Intention
Today we will be learning how matter cycles through Earth's systems, using carbon as our example, so we can explain how carbon moves among the air, oceans, living things, and rocks, and why wetlands like the Everglades matter.

Success Criteria
I can… ● Describe the parts of a system model and explain why matter is conserved while energy drives the cycle. ● Use a quantitative model to describe how carbon moves among the atmosphere, hydrosphere, biosphere, and geosphere, and compare fast and slow carbon cycling. ● Explain how a change in one part of a system can cause a feedback.

The Everglades: A River of Grass

Everglades National Park, in southern Florida, protects the largest subtropical wilderness in the United States. Much of it is a shallow, slow-moving sheet of water flowing through sawgrass, which is why the Everglades is called a "River of Grass." For thousands of years, dead sawgrass and other plants have piled up in the water and turned into a thick, spongy soil called peat. That peat holds a large amount of carbon that the plants once pulled from the air. In this guide, the Everglades is our example of how carbon is stored and moved in Earth's systems. As you read, keep this question in mind: *Where does carbon go, and what happens when one part of the system changes?*

Systems and System Models

A **system** is a group of parts that work together. Scientists use **system models** to show how matter and energy move through a system. A system model has:

- **Reservoirs:** places where matter is stored, such as the atmosphere, the ocean, living things, or rocks.
- **Flows:** processes that move matter from one reservoir to another, such as evaporation or photosynthesis.
- **Inputs and outputs:** matter or energy entering or leaving the system.

For matter, Earth is almost a closed system: very little matter enters or leaves, so the **total amount of matter is conserved**. It moves from one reservoir to another, as in the rock cycle and the water cycle. For energy, Earth is an open system. **Energy** from the Sun and from Earth's interior enters the system and **drives** the flows, as you learned in the first learning guide.

Check for Understanding

1. Write **R** if the example is a reservoir or **F** if it is a flow.

_____ a. The ocean

_____ b. Photosynthesis

_____ c. The atmosphere

_____ d. A volcano releasing carbon dioxide

_____ 2. Which statement best describes Earth as a system?
(1) Matter and energy both enter and leave freely.
(2) Matter is mostly conserved, while energy enters and leaves.
(3) Energy is conserved, while matter enters and leaves freely.
(4) Neither matter nor energy can move.

The Carbon Cycle: Reservoirs and Flows

Carbon is stored in all four of Earth's spheres. Scientists measure it in **gigatons of carbon (GtC)**. One gigaton is one billion metric tons.

- **Atmosphere:** carbon dioxide gas (CO_2) in the air.
- **Hydrosphere:** carbon dissolved in ocean water. The ocean holds much more carbon than the air.
- **Biosphere:** carbon in living things, dead plants, and soil, including peat.
- **Geosphere:** carbon in rocks, mostly limestone, and in fossil fuels such as coal, oil, and natural gas. This is by far the largest reservoir.

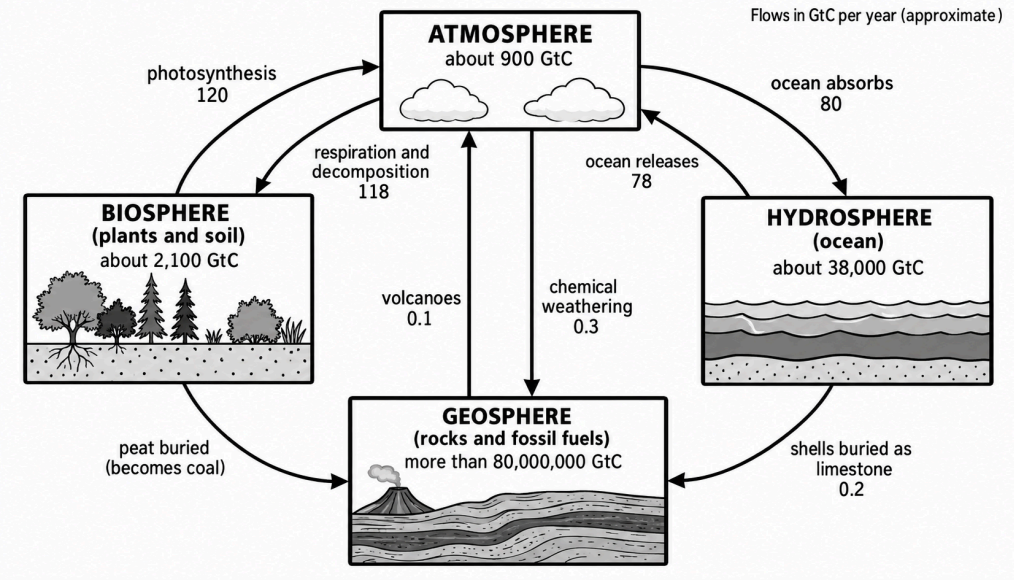
Carbon moves among these reservoirs through these processes:

- **Photosynthesis:** plants take CO_2 from the air to build their bodies (atmosphere → biosphere).
- **Respiration and decomposition:** living things and decomposers release CO_2 back to the air (biosphere → atmosphere).
- **Ocean exchange:** the ocean absorbs CO_2 from the air and releases it again (atmosphere ↔ hydrosphere).
- **Burial:** shells settle on the sea floor and become limestone, and buried plants become peat and later coal (hydrosphere and biosphere → geosphere).
- **Chemical weathering:** CO_2 dissolves in rain to form a weak acid that dissolves rock. Rivers carry the dissolved material to the ocean, where it is eventually locked into limestone (atmosphere → geosphere).
- **Volcanic outgassing:** volcanoes release carbon stored deep in Earth (geosphere → atmosphere). Burning fossil fuels also moves carbon from the geosphere to the atmosphere. You will study this in Unit 11.

Some flows are fast and huge, such as photosynthesis and ocean exchange, which move carbon in years to centuries. Others, such as chemical weathering, burial, and volcanic outgassing, are slow and small each year but move carbon over **millions of years**. This is the difference between the **fast carbon cycle** and the **slow carbon cycle**.

Check for Understanding: Model It

The model shows the approximate amount of carbon in each reservoir (GtC) and the approximate amount that moves each year (GtC per year).



_____ 1. Which reservoir stores the most carbon?
(1) atmosphere
(2) biosphere
(3) hydrosphere
(4) geosphere

2. Use the model to calculate the net amount of carbon that moves from the atmosphere into the biosphere each year. Show your work.

_____ GtC per year

_____ 3. Which two flows are part of the slow carbon cycle?
(1) photosynthesis and respiration
(2) ocean absorbs and ocean releases
(3) chemical weathering and volcanoes
(4) photosynthesis and ocean absorbs

4. Explain why the flows of the slow carbon cycle are so small each year, yet they control the largest reservoir of carbon on Earth.

_____

_____

Check for Understanding: Fast vs. Slow

Scientists also measure how long carbon usually stays in a reservoir before a flow moves it somewhere else. This is called the carbon's **average time in the reservoir**. The table compares the amount of carbon in each reservoir with how long carbon usually stays there. Values are approximate.

Reservoir	Approximate carbon stored (GtC)	Average time carbon stays
Atmosphere	900	about 5 years
Biosphere (plants and soil)	2,100	about 10 to 100 years
Hydrosphere (ocean)	38,000	hundreds to about 1,000 years
Geosphere (rocks)	more than 60,000,000	more than 100 million years

_____ 1. Which pattern does the table show?
(1) Larger reservoirs usually hold carbon for a longer time.
(2) Larger reservoirs usually hold carbon for a shorter time.
(3) All reservoirs hold carbon for about the same time.
(4) The atmosphere holds the most carbon for the longest time.

2. Use the table to explain why carbon stored in limestone is part of the slow carbon cycle, while carbon in the atmosphere is part of the fast carbon cycle.

_____

_____

The Everglades as a Carbon Store

In the Everglades, sawgrass takes in CO_2 through photosynthesis. When the plants die, they fall into water that has very little oxygen, so decomposers break them down very slowly. The plant material piles up faster than it decays and forms peat. Over thousands of years, the Everglades built up layers of peat that store a large amount of carbon. If peat is buried and compacted for millions of years, it can become coal.

In the early 1900s, people began draining parts of the Everglades for farms and cities. Once the peat was exposed to air, decomposers broke it down much faster, releasing its carbon as CO_2. In some farm areas, the ground has sunk several feet as the peat disappeared.

Draining can also start a **feedback**, a change that causes effects that make the original change bigger. When a wetland shrinks, less water evaporates from it. The local air becomes drier, and less rain falls, so even more of the wetland dries out.

Check for Understanding

_____ 1. Peat builds up in the Everglades mainly because
(1) plants grow only in winter
(2) waterlogged soil slows decomposition
(3) volcanoes add carbon to the water
(4) the ocean deposits shells on land

2. **Model It.** Complete the feedback loop for a shrinking wetland.

Wetland shrinks → Less water evaporates → _____

→ _____ → Wetland shrinks even more

Check for Understanding: Trace a Carbon Atom

A single carbon atom can travel through all four spheres. Follow one atom's journey below. For each step, name the process and write **Fast** or **Slow**. The first step is done for you.

Step	The carbon atom moves…	Process	Fast or Slow?
A	from CO_2 in the air into a sawgrass leaf	Photosynthesis	Fast
B	from drained, dried-out peat back into the air		
C	from the air into ocean water		
D	from ocean water into a shell that becomes limestone		
E	from rock deep in Earth back into the air through a volcano		

_____ 1. During steps A through E, the carbon atom
(1) is destroyed and replaced by a new atom
(2) changes into a different element
(3) stays the same atom as it moves between reservoirs
(4) leaves Earth and returns from space

2. Which step (A, B, C, D, or E) would take the longest to happen? Explain your choice.

_____

_____

Cluster Set 1

Base your answers to questions 1 through 4 on the data table below and on your knowledge of Earth and Space Science.

The table shows a simplified model of the carbon that enters and leaves the atmosphere each year. Values are approximate.

Carbon added to the atmosphere	GtC per year	Carbon removed from the atmosphere	GtC per year
Respiration and decomposition	118	Photosynthesis	120
Ocean releases	78	Ocean absorbs	80
Volcanoes	0.1	Chemical weathering	0.3
Burning fossil fuels	10		

1. Calculate the total amount of carbon added to the atmosphere each year.

_____ GtC per year

2. Calculate the total amount of carbon removed from the atmosphere each year.

_____ GtC per year

_____ 3. Based on your answers, the amount of carbon in the atmosphere is
(1) decreasing by about 6 GtC each year
(2) staying the same each year
(3) increasing by about 6 GtC each year
(4) increasing by about 200 GtC each year

4. Which single flow in the table is most responsible for the change you found in question 3? Use the data to explain your answer.

_____

_____

Mini Glossary

Chemical weathering – the breakdown of rock by chemical reactions, such as rain containing dissolved CO_2

Decomposition – the breakdown of dead organisms by decomposers, which releases CO_2

Fast carbon cycle – carbon flows that happen in years to centuries, such as photosynthesis and ocean exchange

Feedback – a change that causes effects that make the original change bigger or smaller

Flow – a process that moves matter from one reservoir to another

Gigaton of carbon (GtC) – one billion metric tons of carbon

Peat – a spongy soil made of partly decayed plants in waterlogged wetlands

Photosynthesis – the process plants use to take in CO_2 and make food

Reservoir – a place where matter is stored in a system

Respiration – the process living things use to release energy, giving off CO_2

Slow carbon cycle – carbon flows that happen over millions of years, such as weathering, burial, and volcanoes

System – a group of parts that work together

Volcanic outgassing – the release of gases, including CO_2, from Earth's interior by volcanoes